

Bachelor in Computer Applications

Course Title: **Microprocessor and Computer Architecture**

Code No: Semester: **2**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. What is 8085 microprocessor? Explain the functional units of 8085 microprocessor in brief.
2. Draw the block diagram of Hardwired control unit and explain their functional units in brief.
3. State an algorithm for the restoring division method of fixed point binary division. Show the step-by step division process using restoring division algorithm when 448(01110 00000) is divided by 17(10001). Use 5-bit register to represent the numbers.
4. What is the difference between SAP-1 and SAP-2?
5. What is instruction cycle? If the clock frequency is 5 MHz, how much time is required to execute an instruction MVI A, 08H?
6. What is Microoperations? Explain the function of each following microoperations $AR \leftarrow PC$, $DR \leftarrow M[AR]$, $PC \leftarrow PC+1$, $AR \leftarrow DR(0 - 10)$, $CAR(2 - 5) \leftarrow DR(11 - 14)$
7. What is Instruction Format of basic computer architecture? A computer uses a memory unit with 256K words of 32 bits each. Draw the instruction word format and indicate the number of bits in each part.
8. Write a 8085 assembly language for division of two 8-bit numbers by repeated subtraction method.
9. Explain the use of mapping table.
10. What is the difference between isolated I/O and memory-mapped I/O? What are the advantages and disadvantages of each?
11. What is pipelining? Explain the role of pipelining in computing.
12. Write short notes on:
a. Multiplexed Address
b. Memory Hierarchy